

GRADWAVE: a differentiable plane-wave DFT code in PyTorch

Capability portfolio: performance and automatic differentiation

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Abstract

GRADWAVE is a plane-wave pseudopotential DFT code built on PyTorch, designed so that every physical quantity is a differentiable function of its inputs. It reproduces QUANTUM ESPRESSO 7.5 at mature-code precision (equation-of-state Δ -factor ≤ 0.17 meV/atom over six materials; the full QE-comparison test suite passes on both CPU and a consumer GPU), while exposing capabilities that conventional codes cannot: exact gradients of the total energy with respect to exchange–correlation parameters, Hubbard U , and atomic positions, obtained from the computation graph rather than hand-derived matrix elements. This document collects a wall-clock comparison against QUANTUM ESPRESSO and four automatic-differentiation showcases: learnable exchange–correlation functionals, the Hubbard U as a differentiable and self-determinable parameter, phonons from autograd forces, and noncollinear magnetism with the exchange field generated by autograd.

1 Overview and validation summary

GRADWAVE implements norm-conserving pseudopotential DFT (UPF v2, ONCV/PseudoDojo) with LDA/PBE and collinear/noncollinear spin including spin–orbit coupling, in float64/complex128 throughout. The code is organized in three layers: a pure differentiable tensor core (energies, Hamiltonian applies), a setup/solver layer (Davidson, SCF mixing) that runs under `torch.no_grad()`, and a user layer (YAML/CLI, ASE calculator). Autograd never traces the SCF loop; derivatives are obtained at the converged fixed point via stationarity (Hellmann–Feynman), implicit differentiation (Sternheimer response with the Hxc kernel as an autograd Hessian-vector product), or explicit graphs over the position-dependent terms.

The complete integration suite (46 tests, including every slow QE comparison) passes unmodified on CUDA; a rattled-Si₈ NVE trajectory driven by the ASE calculator conserves total energy to a secular drift of ~ 1 μ eV/atom over 0.5 ps — direct evidence that the autograd Hellmann–Feynman force is the *exact* gradient of the SCF energy surface.

2 Performance versus Quantum ESPRESSO

Wall-clock comparison on a single machine (Intel laptop, 22 hardware threads, NVIDIA RTX 3050 6 GB). QUANTUM ESPRESSO 7.5 runs with `mpirun -np 8` (its efficient parallelization sweet spot for these small cells); GRADWAVE runs on 22 CPU threads or on the GPU. Both codes use identical plane-wave cutoffs, k -meshes (symmetry reduction on in both), FFT grids (pinned), band counts, functionals, and comparable convergence thresholds (`conv_thr` = 10^{-10} Ry vs `etol` = 10^{-9} eV). Times are full SCF wall times excluding setup.

Table 1: Headline agreement with QUANTUM ESPRESSO 7.5 at identical pseudopotentials, cut-offs, k -meshes, and FFT grids.

benchmark	quantity	GRADWAVE vs QUANTUM ESPRESSO	scale
SCF energies (Si, Al, Cu, Fe, GaAs, ...)	E or F	$\leq 2 \text{ meV/atom}$ (most $\leq 10^{-3}$)	—
Δ -factor EOS (6 materials, $\pm 6\%$)	Δ	$\leq 0.17 \text{ meV/atom}$	matur
lattice constants from EOS	a_0	$\leq 4 \times 10^{-4} \text{ \AA}$	—
forces (displaced Si, Al)	F	$\leq 5 \text{ meV/\AA}$	—
NVE drift (Si ₈ , 0.5 ps, autograd forces)	E_{tot} secular	$\sim 1 \mu\text{eV/atom}$	exact
dielectric constant (Si)	ϵ_{∞}	0.002 %	vs ph
Born charges (MgO)	Z^*	0.07–0.10 %	vs ph
band structures (Si)	$\epsilon_{n\mathbf{k}}$	$\leq 10 \text{ meV}$	—
SOC splitting (GaAs Δ_0)	0.3360 vs 0.3360 eV	0.03 meV	spino
NiO DFT+ U (AFM-II)	E_U	1.6 meV	occup

Table 2: SCF wall time (seconds); identical physics settings, energies agree to $\lesssim 1 \text{ meV/atom}$. Iteration counts in parentheses.

system	N_{at}	ecut (Ry)	k -mesh	QUANTUM ESPRESSO (8 MPI)	GRADWAVE CPU-22	GRADWAVE GP
Si (dia.)	2	30	4^3	1.5 (6)	3.5 (10)	1.5 (1)
C (dia.)	2	40	4^3	1.5 (6)	1.8 (11)	0.9 (1)
GaAs	2	40	4^3	1.8 (9)	7.9 (14)	2.8 (1)
Al (fcc)	1	40	8^3	1.8 (6)	5.1 (11)	4.2 (1)
Cu (fcc)	1	45	8^3	2.2 (9)	10.0 (13)	5.1 (1)
MgO	2	50	4^3	1.6 (9)	3.6 (13)	1.6 (1)
Si ₈	8	30	2^3	1.9 (8)	5.8 (13)	5.4 (1)

The honest summary: on cells this small, QUANTUM ESPRESSO’s two decades of MPI engineering show — it is on par with or faster than GRADWAVE’s GPU path and 2–5 $\times$ faster than the 22-thread CPU path. Two structural factors account for most of the gap: QUANTUM ESPRESSO converges in 6–9 SCF iterations to GRADWAVE’s 10–14 (mixer maturity, not tensor throughput — per-iteration GPU cost is at or below QUANTUM ESPRESSO’s), and at these sizes (≤ 3000 plane waves) kernel-launch and Python overhead are proportionally largest. The GPU advantage grows with system size (e.g. a 78-electron Bi₂Se₃ SOC calculation on an 80^3 grid runs its SCF in minutes on the 6 GB card); the relevant point for this portfolio is that the fully differentiable formulation costs little enough that the same code base is usable for production ground states *and* everything in the following sections.

3 Differentiable exchange–correlation functionals

The XC functional is an `nn.Module`; $v_{\text{xc}} = \delta E_{\text{xc}}/\delta \rho$ is generated by autograd (including GGA gradient terms), so any twice-differentiable functional — including one with trainable parameters — is a drop-in. At SCF convergence the energy is variational in the density, so the training gradient is free:

$$\frac{dE}{d\theta} = \left. \frac{\partial E_{\text{xc}}[\rho; \theta]}{\partial \theta} \right|_{\rho^*}, \quad (1)$$

one backward pass, no response solve, no finite differences. Table 3 verifies this against central finite differences of full SCF re-runs for a PBE-form exchange with learnable (κ, μ) (initialization at PBE values reproduces PBE to 10^{-12}).

Table 3: $dE/d\theta$ for learnable exchange parameters (Si, off-PBE point $\kappa = 0.65$, $\mu = 0.18$): one autograd backward vs. SCF finite differences.

parameter	autodiff	finite difference	diff
θ_κ	-0.1911153641	-0.1911153645	3.3×10^{-10}
θ_μ	-1.6478174399	-1.6478174406	6.7×10^{-10}

As an end-to-end demonstration, starting from an off-PBE point $(\kappa, \mu) = (0.55, 0.30)$ and fitting to PBE total energies at two volumes with Adam:

loss $6.5 \times 10^{-1} \rightarrow 3.8 \times 10^{-4}$ in five steps; $\mu \rightarrow 0.2222$ vs PBE’s 0.2195 after ten steps.

The energy data pin down μ (in silicon’s reduced-gradient range, $F \approx 1 + \mu s^2$ to leading order) while κ — an s^4 effect — is left nearly unconstrained and wanders a flat valley: the optimization itself performs a physically meaningful identifiability analysis of the functional form. Density-based losses use the implicit SCF backward (self-consistent adjoint/Sternheimer, validated to 2×10^{-4} against finite differences).

4 The Hubbard U as a differentiable quantity

GRADWAVE implements rotationally invariant (Dudarev) DFT+ U with QE’s `atomic` projector convention, validated on AFM-II NiO against QUANTUM ESPRESSO (E_U : 2.9956 vs 2.9942 eV; Ni- d occupations 4.909/3.647 vs 4.910/3.648; $E(U=5) - E(U=0)$: 3.3245 vs 3.3216 eV). On top of this, three derivative capabilities:

Exact dE/dU . By stationarity, $dE_{\text{tot}}/dU = \sum_{I\sigma} \frac{1}{2} \text{Tr}[n^{I\sigma}(1 - n^{I\sigma})]$ at the converged point — the gradient a U -learning loop would backpropagate. NiO at $U = 5$ eV: analytic 0.616154 vs SCF finite differences 0.616153 (|diff| = 7.4×10^{-7}).

+ U forces by autograd. E_U depends on positions only through the projector phases $e^{-i(\mathbf{k}+\mathbf{G})\cdot\tau}$; autograd through the phase factors gives $-dE_U/d\tau$ directly. On symmetry-broken NiO the analytic force matches frozen-orbital finite differences to 4.9×10^{-8} eV/Å and vanishes identically on atoms without a correlated manifold.

The code computes its own U . The Cococcioni–de Gironcoli linear-response $U = (\chi_0^{-1} - \chi^{-1})_{II}$ is implemented two ways. The finite-difference route ($\pm\alpha$ probe SCF re-runs) gives $U = 6.4493$ eV for NiO. The *analytic* route obtains both response matrices from a single ground state: a batched conduction-projected Sternheimer solve with the projector probe on the right-hand side, screened self-consistently through the spin Hxc kernel evaluated as an autograd Hessian-vector product of E_{Hxc} — DFPT without hand-coding $f_{\text{xc}}^{\sigma\sigma'}$, valid for any twice-differentiable spin functional including learnable ones. Anderson acceleration is essential: the antisymmetric magnetization mode across the two Ni sites gives $K\chi_0$ an eigenvalue ≈ -6 (PBE NiO is near a spin instability), which diverges under plain damped iteration; Anderson converges in 11 iterations.

The two GRADWAVE routes agree to 0.3 meV (and column-by-column to 2×10^{-5} in χ , χ_0); both agree with QUANTUM ESPRESSO’s independent DFPT implementation to 0.3%.

Table 4: Linear-response Hubbard U for NiO (Ni $3d$, PBE, PseudoDojo, 40 Ry, 2^3 k -mesh).

method	U (eV)
GRADWAVE, finite-difference response ($\pm\alpha$ SCFs)	6.4493
GRADWAVE, analytic Sternheimer + autograd-HVP kernel	6.4496
QUANTUM ESPRESSO <code>hp.x</code> (DFPT, $n_q = 1$)	6.4308

5 Phonons from autograd forces

Forces are exact autograd gradients, so force-constant matrices from finite displacements inherit machine-grade consistency: the Hessian entry from displaced *forces* matches the second difference of *energies* to $< 0.5\%$, acoustic modes vanish under the sum rule, and the NVE test of Sec. 1 bounds any systematic force error at the μeV scale. At settings identical to a QUANTUM ESPRESSO `ph.x` DFPT calculation (PBE, 40 Ry, 4^3 mesh, pinned 32^3 grid):

Table 5: Si Γ phonons (cm^{-1}): finite differences of GRADWAVE autograd forces vs QUANTUM ESPRESSO `ph.x` DFPT.

mode	GRADWAVE (FD of autograd forces)	QUANTUM ESPRESSO <code>ph.x</code> (DFPT)
acoustic T_{1u} ($\times 3$)	0.000 (ASR)	6.03 (raw, no ASR)
optical T_{2g} ($\times 3$)	522.55	523.35

The optical frequency agrees with the independent DFPT result to 0.8 cm^{-1} (0.15%), the triplet is degenerate to the printed precision, and the acoustic modes vanish exactly under the sum rule — all from displacements of $5 \times 10^{-3} \text{ \AA}$ evaluated with autograd forces, with no analytic force implementation anywhere in the code.

For polar materials the ingredients for LO–TO splitting are available from the same response stack: ϵ_∞ and Born charges (Sec. 1) are computed by E-field DFPT in which the Born charge is literally a mixed derivative $\partial^2 E / \partial \mathcal{E} \partial \tau$ — one autograd backward through the position-differentiable pseudopotential per field direction yields Z^* for all atoms simultaneously.

6 Noncollinear magnetism with autograd exchange fields

The noncollinear (spinor) SCF uses locally collinear XC in which a *single* autograd call on $E_{\text{xc}}[\rho, |\mathbf{m}|]$ yields both v_{xc} and the exchange field $\mathbf{B}_{\text{xc}} \parallel \mathbf{m}$ — no hand-derived functional derivatives of the spin rotation. Validation ladder on bcc Fe:

- collinear limit: spinor SCF with $\mathbf{m} \parallel \hat{z}$ reproduces the two-channel calculation to 0.0000 meV;
- rotational invariance: total energies for $\mathbf{m}$ along $\hat{x}$, $\hat{z}$, and 45° agree to $0.19 \mu\text{eV}$ (no SOC — the exact symmetry emerges numerically from the autograd field);
- moment directions are preserved self-consistently (no drift to a collinear attractor).

With fully relativistic j -resolved projectors (spin–orbit coupling built from complex spherical harmonics and Clebsch–Gordan spin structure on the doubled plane-wave axis):

- GaAs: free energy matches QUANTUM ESPRESSO `lspinorb` to 0.0007 meV/atom; split-off gap $\Delta_0 = 0.3360$ vs 0.3360 eV (0.03 meV);
- Bi₂Se₃: the topological band inversion is reproduced — the Γ gap-edge parities swap between scalar-relativistic and SOC calculations, with parity eigenvalues ± 1.000 and exact Kramers degeneracy (78 electrons, 80^3 grid, NLCC, on the RTX 3050);
- magnetic energy mapping: bcc Fe Heisenberg couplings $J_1 = +57.5$ meV, $J_2 = -39.6$ meV (mean-field $T_c \approx 860$ K) from signed-shell energy differences; Cr correctly orders AFM (-27.7 meV/atom vs nonmagnetic, matching QUANTUM ESPRESSO to 0.02 meV/atom).

The same differentiable structure is what makes magnetic *response* quantities tractable next: the spin Hxc kernel used for the Hubbard- U response (Sec. 4) is the 2×2 spin block of exactly this autograd machinery, and moment-direction torques $dE/d\hat{\mathbf{m}}$ for magnetocrystalline anisotropy follow the same pattern with SOC.

7 Closing remarks

Every result above ran on commodity hardware (a laptop CPU and a 6 GB consumer GPU) against reference calculations from QUANTUM ESPRESSO 7.5 with identical numerical settings. The differentiable formulation did not cost ground-state precision (Sec. 1) or speed (Sec. 2); it added a family of exact derivatives — $dE/d\theta_{xc}$, dE/dU , $dE/d\tau$, $\partial^2 E/\partial\mathcal{E}\partial\tau$, self-consistent response via autograd Hxc kernels — that conventional codes obtain only through hand-derived, functional-specific implementations, if at all.